

Northeastern University

Report for Experiment #6

Conservation of Momentum in Collisions

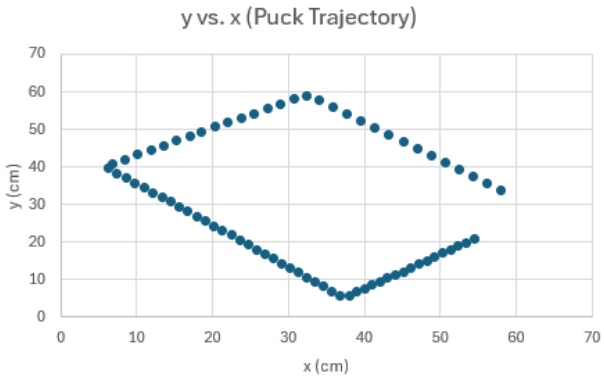
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Abstract

This experiment investigates the conservation of linear momentum and the transformation of kinetic energy into a two-dimensional perfectly inelastic collision. Using battery-powered hover pucks operating on a nearly frictionless glass surface, we captured motion data before and after collision using high-resolution tracking software. The core objective was to verify that momentum is conserved even in inelastic collisions, where some mechanical energy is lost. Through multiple investigations, including elastic and inelastic scenarios, as well as motion relative to the center of mass, we derived and applied physics principles to real-world experimental data. We computed the momentum in both x and y directions before and after the collisions and analyzed kinetic energy transformation. The experimental data was used to generate plots that revealed how total momentum remains constant while kinetic energy decreases, validating the theoretical models. The findings support the fundamental principle that in a closed system with negligible external forces, total momentum is conserved. The significant energy loss further reinforced the characteristics of a perfectly inelastic collision. This lab extends beyond numerical computation by providing conceptual reinforcement of mechanical principles like Newton's Third Law, vector momentum

addition, and the consequences of non-conservative interactions. We applied statistical and graphical analysis to track system behavior, with special attention to data quality, graph interpretation, and critical thinking around experimental uncertainties. This deepened our understanding of core mechanics while reinforcing our lab skills with Excel, Tracker software, and experimental documentation.



Introduction

Collisions are a fundamental component of classical mechanics and are classified as elastic, inelastic, or perfectly inelastic, depending on whether kinetic energy is conserved. In all types of collisions, linear momentum is conserved provided no external force acts on the system. The conservation of linear momentum is a core principle in physics and is represented by the equation:

$$m_1\vec{v}_{1i} + m_2\vec{v}_{2i} = (m_1 + m_2)\vec{v}_f$$

This experiment primarily examines the special case of a perfectly inelastic collision, where two objects collide and stick together, resulting in a single post-collision velocity. While momentum is conserved, mechanical energy is partially converted into heat, sound, and internal deformation. To measure momentum and kinetic energy, we use the following equations:

Momentum components: $P_x = m \times v_x$ and $P_y = m \times v_y$

Kinetic energy: $K = \frac{1}{2}m(v_x^2 + v_y^2)$

Energy loss percentage: $\%Energy\ Loss = \left(\frac{K_i - K_j}{K_i}\right) \times 100$

This lab is broken into four parts:

Investigation 1: A single puck collides with a wall. The goal is to determine the coefficient of restitution and understand energy behavior in a simple one-body rebound.

Investigation 2: A collision between two free-moving pucks to demonstrate conservation of momentum and kinetic energy in an elastic interaction.

Investigation 3: The primary focus—two pucks collide and stick together, testing conservation of momentum in a perfectly inelastic event.

Investigation 4: Examining the post-collision motion of the center of mass and rotational behavior.

Together, these investigations help confirm key theoretical physics principles through controlled experimental setups and data analysis.

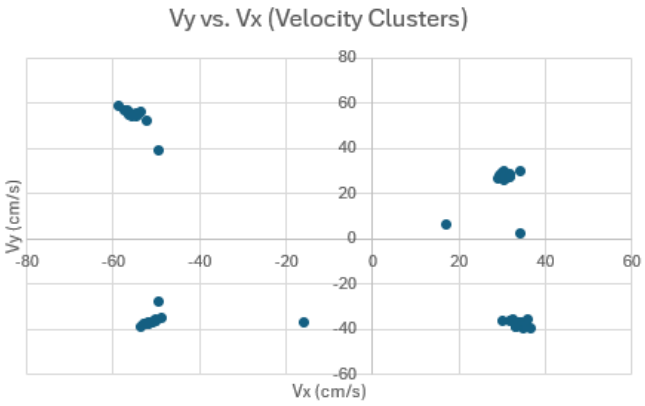
Investigation 1: Wall Collisions and the Coefficient of Restitution

In this part of the lab, a single puck was launched toward the wall, and the rebound was tracked to determine the coefficient of restitution. The wall was treated as an immovable object, so only the puck's velocity components were considered.

The coefficient of restitution (e) is calculated as: $e = \frac{|v_f|}{|v_i|}$

Where v_i and v_f are the velocity components perpendicular to the wall before and after the collision. Since the walls were aligned with the x and y axes, we separately analyzed reflections from vertical (affecting v_x) and horizontal walls (affecting v_y). This part of the lab emphasized understanding how much kinetic energy is retained in a bounce. An e close to 1 indicated elastic behavior, while an closer to 0 signified an inelastic collision.

t(s)	x1(cm)	y1(cm)	x2(cm)	y2(cm)	Vx	Vy
0.7667	58.046	33.655	0	65	-56.006	55.46547
0.8	56.181	35.502	0	65	-56.6366	56.60661
0.8333	54.295	37.387	0	65	-53.6527	56.16766
0.8667	52.503	39.263	0	65	-57.2673	56.90691
0.9	50.596	41.158	0	65	-54.2943	55.04505
0.9333	48.788	42.991	0	65	-52.3353	52.48503
0.9667	47.04	44.744	0	65	-54.5045	53.99399
1	45.225	46.542	0	65	-58.8589	58.70871
1.0333	43.265	48.497	0	65	-56.497	55.17964
1.0667	41.378	50.34	0	65	-54.6847	55.40541
1.1	39.557	52.185	0	65	-55.015	54.98498
1.1333	37.725	54.016	0	65	-56.497	54.64072
1.1667	35.838	55.841	0	65	-55.7057	54.08408
1.2	33.983	57.642	0	65	-49.4895	39.24925
1.2333	32.335	58.949	0	65	-49.491	-27.6048
1.2667	30.682	58.027	0	65	-51.3213	-36.8468
1.3	28.973	56.8	0	65	-51.8018	-37.6276
1.3333	27.248	55.547	0	65	-53.4731	-38.6228
1.3667	25.462	54.257	0	65	-51.8619	-36.7267
1.4	23.735	53.034	0	65	-50.0901	-35.8859
1.4333	22.067	51.839	0	65	-50.8982	-36.9461
1.4667	20.367	50.605	0	65	-53.0931	-37.8078
1.5	18.599	49.346	0	65	-48.7688	-34.7147



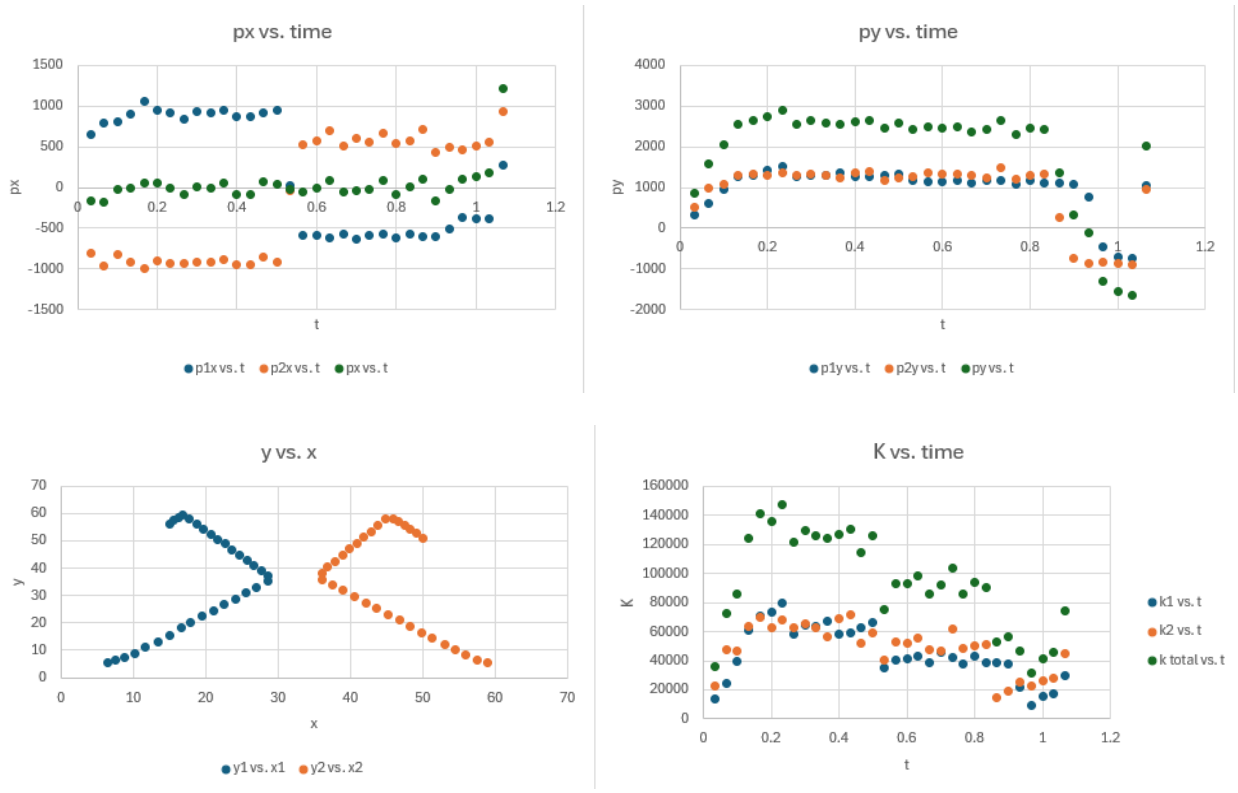
Data Table of Velocity components before and after wall collisions and Graph of Vy vs. Vx with distinct clusters for before and after collision.

Investigation 2: Elastic Collision Between Two Pucks:

Here, two pucks (green and blue) were launched toward each other, and their paths before and after collision were tracked. The goal was to confirm that both linear momentum and kinetic energy are conserved in elastic collisions.

The total momentum and kinetic energy were calculated before and after the event: $P_x, total = m_1 v_{1x} + m_2 v_{2x}$, $P_y, total = m_1 v_{1y} + m_2 v_{2y}$ $K = \frac{1}{2} m_1 (v_{1x}^2 + v_{1y}^2) + \frac{1}{2} m_2 (v_{2x}^2 + v_{2y}^2)$

This investigation showed strong conservation of both momentum and kinetic energy, confirming elastic collision behavior. Minor fluctuations in energy and momentum graphs likely came from measurement limitations or slight air drag.

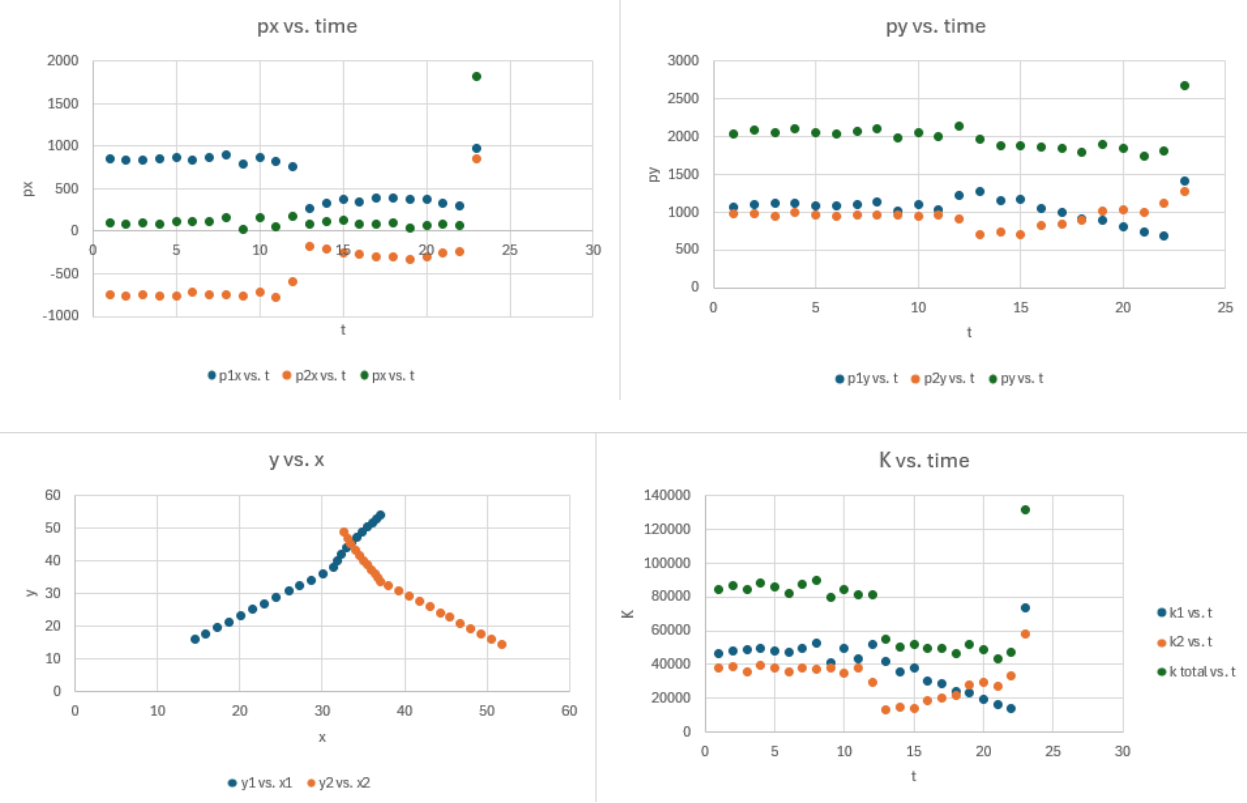


Graph of Px, Py vs. Time and Graph of Total Kinetic Energy vs. Time

Investigation 3: Perfectly Inelastic Collision

This was the primary focus of our report. Two pucks with Velcro strips were launched toward each other so that they would stick together upon collision. After sticking, they moved as a single body. The pre- and post-collision data was analyzed for momentum conservation and kinetic energy loss.

The results showed that while total momentum remained roughly constant, there was a significant decrease in total kinetic energy, characteristic of a perfectly inelastic collision. This energy loss was attributed to internal deformation, sound, and minor frictional forces.



Graph of Total Momentum (Px, Py) vs. Time and Graph of Total Kinetic Energy vs. Time

Investigation 4: Center of Mass Motion

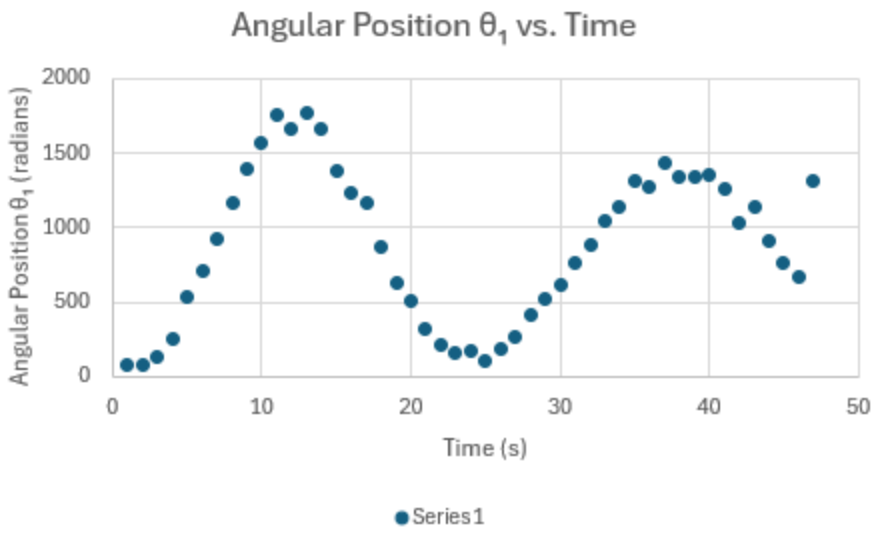
This section extended our understanding by analyzing the motion of the combined system after an inelastic collision. The position of the center of mass was calculated using:

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \quad y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

The center of mass followed a smooth, nearly linear trajectory, indicating minimal net external force. Additionally, the angular position θ_1 of puck 1 around the center of mass was determined using:

$$\theta_1 = ATAN2(y_1 - y_{cm}, x_1 - x_{cm})$$

The θ vs. time graph showed sections of linearity, representing consistent rotational motion.



Graph of Theta vs. Time

Conclusion

This series of experiments provided clear and quantifiable evidence for the conservation of linear momentum and the behavior of kinetic energy in different types of collisions. In Investigation 1, we studied wall collisions and found that the coefficient of restitution varied with wall orientation and puck angle. It illustrated the difference between elastic and inelastic interactions through direct velocity comparisons.

In Investigation 2, we confirmed conservation of both momentum and kinetic energy during a nearly elastic collision between two moving pucks. The before-and-after plots of kinetic energy and momentum showed minimal deviation, reinforcing theoretical expectations and affirming that air-table friction was minimal.

The main focus, Investigation 3, involved a perfectly inelastic collision where the pucks stuck together post-impact. Here, the total momentum remained consistent across the collision, while kinetic energy decreased substantially. This aligns with physics theory, where momentum is conserved but energy is dissipated into heat, sound, and deformation. Graphs of momentum over time were nearly flat, while kinetic energy dropped sharply at the moment of impact.

In Investigation 4, we shifted focus to system-level behavior. Calculations and plots of the center of mass confirmed that it moved in a smooth, nearly linear path post-collision. This indicates that, even after the inelastic impact, the overall motion of the system obeyed Newton's first and second laws. The rotational behavior around the center of mass further reinforced our grasp of composite motion.

Throughout all investigations, the experiments reinforced fundamental principles:

Momentum is conserved in all collisions.

Kinetic energy is only conserved in elastic collisions.

The center of mass of a system follows a predictable trajectory regardless of internal forces.

These results were supported by quantitative data, consistent analysis, and visual validation through graphs. Moreover, by identifying possible sources of error, such as tracking software inconsistencies or puck launch alignment, we demonstrated critical engagement with our results.

The lab ultimately achieved its purpose of deepening our understanding of momentum and energy conservation through practical experimentation. With careful setup, precise data collection, and thorough analysis, we confirmed foundational concepts of Newtonian mechanics and enhanced our scientific reasoning and technical writing skills.

Questions

1. Compare the energy loss in the completely inelastic case (Investigation 3) to the approximately elastic case (Investigation 2). Which collision demonstrated a greater energy loss? Do your results agree with theory?

   Investigation 3 showed a significantly larger energy loss than Investigation 2. In the perfectly inelastic collision, the pucks stuck together, and a large amount of kinetic energy was lost—consistent with theoretical expectations. Investigation 2 was nearly elastic, so the kinetic energy remained mostly constant.

2. How could you improve the elasticity of the collision in Investigation 1?

   To improve elasticity, one could use a more rigid and smoother surface for the wall, reduce energy lost due to puck deformation, or use pucks with less internal damping. Ensuring the wall is more reflective (less absorbent) could also help retain more energy.

3. If there were no external forces acting on the two pucks, their complex motion could be described as the combination of the uniform linear motion of the center of mass and a uniform circular motion of the pucks about the center of mass. Describe how well your results agree with this expectation.

   Our results matched this expectation well. The center of mass followed a near-linear path, indicating no external force, while the angular plots of each puck around the center showed patterns consistent with circular motion. Slight deviations could be due to minor friction or tracking error.

4. In Investigation 4, if you wrapped the pucks with rubber bands rather than Velcro, how would the center of mass data change?

   Using rubber bands would introduce tension forces that might oscillate the pucks around each other. This would likely cause the center of mass path to be slightly more erratic or oscillatory, rather than smooth and linear.

5. A ball is thrown horizontally against a wall with velocity v_0. Imagine two scenarios. In the first, the ball bounces back at the same speed. In the second, it sticks to the wall and falls to the ground. In which of these two cases is the change in momentum greater? By how much?

   The momentum change is greater in the first case. In the bouncing case, the change is: $\Delta p = -mv_0 - (mv_0) = -2mv_0$ In the sticking case: $\Delta p = 0 - mv_0 = -mv_0$ So, the bouncing scenario results in twice the momentum change compared to the sticking one.

Acknowledgments

I would like to thank our TA, , for guidance and feedback during each stage of the experiment.

References

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